SECTION 8 – SHIPPING AND DELIVERY DATA

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8–1 SHIPMENT

Domestic transportation for the MR system, excluding the magnet, will be via an air–ride moving van. The magnet will be shipped on an air–ride flat–bed truck. Export transportation for the MR system overseas will be via air shipment in a pressurized cargo hold. See Table 8–2 for the shipping weights and dimensions of the major MR system components. Actual shipping may vary, international shipment may require equipment to be crated. Refer to Table 8–1 for transportation environmental conditions.

TABLE 8–1
TRANSPORTATION ENVIRONMENTAL CONDITIONS

SYSTEM EQUIPMENT	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	RELATIVE HUMIDITY %	HUMIDITY CHANGE (%/Hr)	ATMOSPHERIC PRESSURE hPa
Electronics Cabinets & equipment	–30 to 140 (–34 to 60)	176 (80)	0–90 non–condensing	30	1012 to 525
Magnet	–31 to 122 (–35 to 50)	176 (80)	0–90 non–condensing	30	1012 to 525

The GE magnet utilizes a Shield/Cryo Cooler System to maintain a reduced helium boil–off. However, the Shield Cooler System is not operational during transportation. Therefore the magnet will require liquid helium replenishment if transportation time exceeds two weeks. Contact GE Service for magnet servicing.

The LCC Magnet is filled with liquid helium at initial shipment but can be allowed to warm up during transportation without damage to the support structure.

8–2 STORAGE REQUIREMENTS

If the system is stored before installation, it must be stored in a warehouse protected from weather. The storage temperature should be between –30 and 140 degrees F (–34 and 60 degrees C) and the relative humidity between 0 and 90% (non–condensing).

There are two scenarios for storing the magnet. One is to maintain the magnet cold temperature by connecting and operating the Shield/Cryo Cooler System. Periodic replenishment of liquid helium may be required depending on the storage time. The other scenario is to store the magnet without maintaining the cold temperature. The LCC magnet can be moved cold or warm, refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation* for magnet moving requirements and details. Contact GE Service for necessary servicing before moving the magnet.

8–3 MAGNET CONSIDERATIONS

For domestic, the magnet is shipped covered with plastic (no shipping pallet). For export, the magnet is crated for shipment on a special shipping pallet. Refer to Table 8–2 for the weight and dimensions of the magnet in its cold ship configuration (i.e. with liquid cryogens in vessel within the cryostat) and with the RF/Gradient Coil inside the magnet bore.

The magnet moving dimensions are shown in Illustration 8–1.

Consideration must be given to the delivery route of the magnet to ensure that the floor can support the magnet and any rigging equipment required to move it. A structural analysis should be performed by a professional structural engineer. The magnet must not be tilted more than 30° in any direction when being moved into position.

The customer is to provide and arrange for riggers to move the magnet from the delivery truck to the final site location. Contact local GE Service for a list of recommended rigger companies. The customer's riggers should have an adequate amount of liability insurance to cover any damage to property or MR system that may occur during delivery of the magnet. The GE sales representative or installation specialist can provide customer riggers with the replacement value of the MR system for insurance purpose.

8–4 COLD–SHIPPED MAGNET DELIVERIES

Cold–shipped magnets are those magnets which are shipped with liquid helium in the vessel within the cryostat. Thus, when these magnets arrive at site, a cryogen delivery route must be available for moving cryogen dewars to the magnet for periodic replenishment of liquid helium. Also, means must be provided for venting of the cryogenic gases if the GE supplied venting kit is not yet installed and if the RF shielded room vent opening is not completed.

Note

Power and cooling water for the shield cooler cabinet must be available when the magnet is delivered to minimize cryogens usage. See Section NO TAG, COOLING REQUIREMENTS, and Section 5, POWER REQUIREMENTS, for specifications.

TABLE 8–2
SHIPPING DATA

MR COMPONENT	APPROXIMATE W x D x H	APPROXI-MATE WEIGHT	METHOD OF SHIPMENT
LCC Magnet with cryogens, partial Wide Open Enclosure installed	93 x 144 x 107 (2362 x 3658 x 2718)	See Note 1	crate/skid
Magnet Accessory Equipment	48 x 48 x 28 (1219 x 1219 x 711)	400 (182)	crate
Shield/Cryo Cooler Compressor Cabinet	26 x 28 x 42 (660 x 711 x 1067)	240 (109)	skid with box cover
Bridge	24 x 161 x 10 (610 x 4090 x 254)	125 (57)	box
Rear Pedestal Assembly	24 x 48 x 36 (610 x 1219 x 914)	40 (18)	box
Enclosure Top	48 x 36 x 36 (1219 x 914 x 914)	30 (14)	box
Enclosure Skirts	40 x 24 x 24 (1016 x 610 x 610)	30 (14)	box
Fixed Site Water Chiller for Gradient Coil Cooling	29 x 42 x 33 (737 x 1067 x 838)	320 (154) dry weight	crate
Patient Table	95 x 36 x 50 (2413 x 914 x 1270)	670 (305)	pallet
1.5T Signa EXCITE II MR/*i* standard & 1.5T Signa MR/*i* 1.5T standard SRFD II Cabinet	24 x 38 x 72 (610 x 965 x 1829)	500 (225)	on cabinet casters, wrapped with plastic
1.5T Signa EXCITE II MR/*i* SRFD II Cabinet with MNS Option	24 x 38 x 72 (610 x 965 x 1829)	565 (256)	on cabinet casters, wrapped with plastic
1.0T SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	500 (227)	on cabinet casters, wrapped with plastic
System Control Cabinet for 1.5T Signa EXCITE MR/*i*	24 x 33 x 78 (610 x 838 x 1981)	475 (215)	on cabinet casters, wrapped with plastic
System Control Cabinet for 1.5T Signa MR/*i* & 1.0T Signa MR/*i*	24 x 33 x 78 (610 x 838 x 1981)	500 (227)	on cabinet casters, wrapped with plastic
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
For 1.5T Signa EXCITE II MR/*i*: GOC Computer Cabinet	24 x 35 x 31 (600 x 900 x 780)	243 (110)	wood pallet with cardboard cover
1.5T Signa MR/*i* & 1.0T Signa MR/*i*: Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Operator Workspace LCD Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	skid
Operator Workspace equipment	32 x 32 x 23 (813 x 813 x 584)	100 (45)	box
Operator Workspace Table	45 x 54 x 37 (1143 x 1372 x 940)	180 (82)	box
BrainWaveHW Lite Cabinet * (GE provided equipment configuration)	24 x 23 x 72 (610 x 584 x 320)	245 (111)	on cabinet casters, wrapped with plastic
MRCC * Outdoor/Indoor Air Cooled:			
MRCC unit	63.8 x 34.3 x 59.1 (1620 x 870 x 1500)	750 (340)	crate
Ship loose items	31.5 x 31.5 x 36 (800 x 800 x 920)	394 (178.5)	box

Note * Optional Equipment

1 Approximate magnet shipping weight (includes packaging material) and weight of magnet with cryogens, RF/Gradient BRM coil, Enclosure parts installed on magnet, lifting beams, crate, and skid (i.e. minus packaging material):
1.0T LCC (Active Shield) 13,200 lbs (6000 kg)
1.5T LCC (Active Shield) 13,900 lbs (6320 kg)

NOTE:

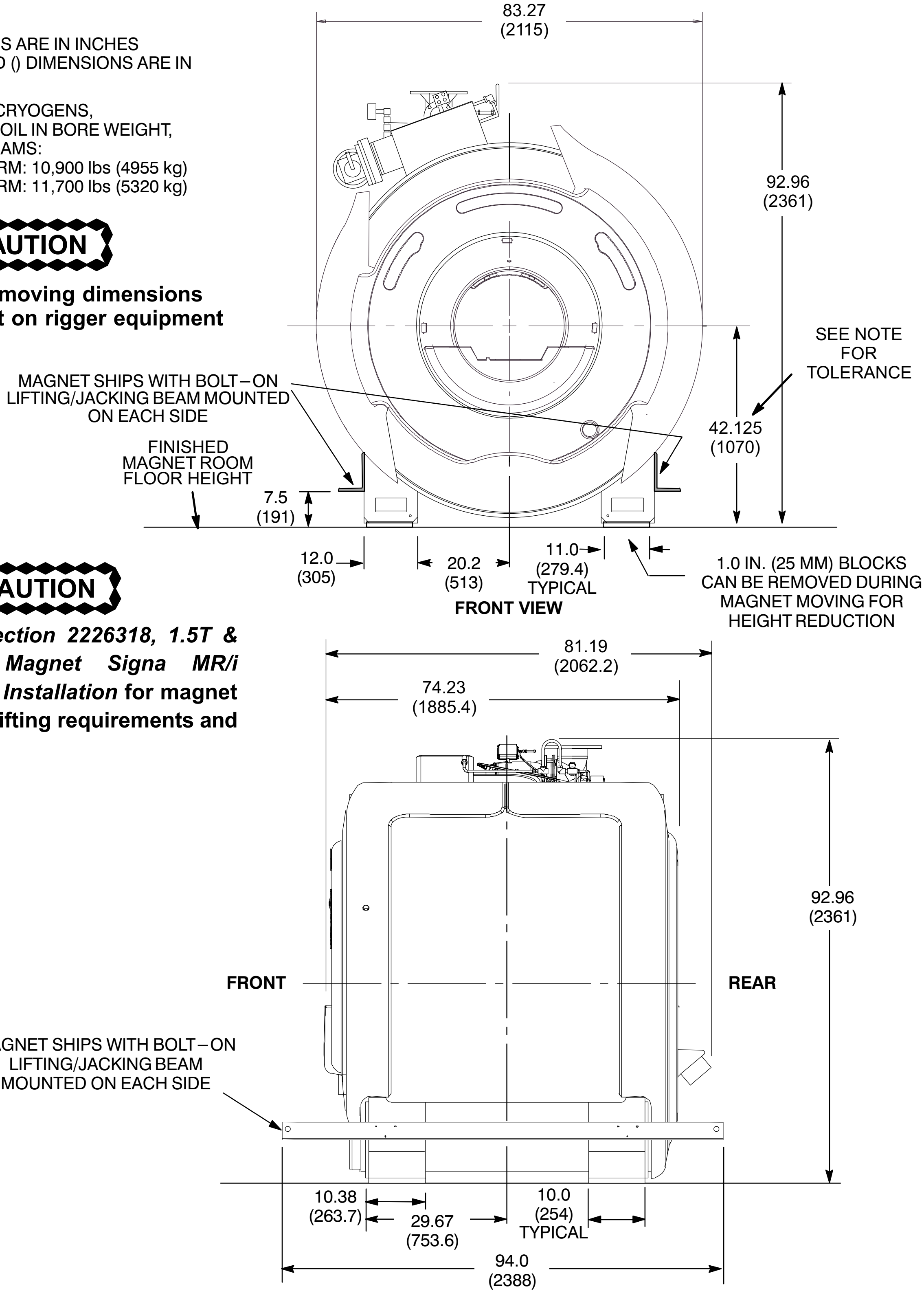
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MAGNET WITH CRYOGENS, RF/GRADIENT COIL IN BORE WEIGHT, AND LIFTING BEAMS:
  For 1.0T WITH BRM: 10,900 lbs (4955 kg)
  For 1.5T WITH BRM: 11,700 lbs (5320 kg)

CAUTION

Final magnet moving dimensions are dependent on rigger equipment requirements.

CAUTION

Refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation* for magnet moving and lifting requirements and details.



1.5T AND 1.0T LCC (ACTIVE SHIELD) MAGNET MOVING DIMENSIONS
ILLUSTRATION 8–1